

Exploring the Impact of Double Taxation Treaties on India's FDI: A Panel Data Approach Using the Gravity Model

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Abstract

Since the liberalization reforms of the early 1990s, India has witnessed a significant surge in foreign direct investment (FDI), positioning itself as a major destination for global capital among emerging economies. FDI has played a crucial role in enhancing industrial productivity, fostering innovation, and accelerating economic growth. Among the key policy instruments aimed at attracting FDI, Double Taxation Avoidance Agreements (DTAAs) are designed to mitigate fiscal barriers, reduce tax uncertainty, and promote cross-border investment. Despite their widespread adoption, the empirical evidence on the effectiveness of DTAAs in driving FDI inflows remains inconclusive, particularly in the Indian context. This study investigates the role of DTAAs in influencing India's bilateral FDI inflows using an augmented gravity model applied to a balanced panel dataset comprising India's 22 major FDI partner countries from 1990 to 2022. The analysis employs Poisson Pseudo Maximum Likelihood (PPML) estimation, supported by robustness checks using Feasible Generalized Least Squares (FGLS) and Newey-West corrected OLS estimators. The findings reveal that the existence of a DTAA significantly enhances FDI inflows to India. Other key determinants include India's GDP, trade openness, FDI openness, and language compatibility, while partner country GDP and colonial ties exert a negative influence. These results underscore the importance of tax treaties and institutional alignment in shaping India's investment climate and provide valuable policy insights for strengthening India's global investment strategy.

1 Introduction

The adoption of liberalization, privatization, and globalization (LPG) policies in the early 1990s marked a transformative shift in India's economic framework. These reforms played a pivotal role in transitioning the country from a tightly regulated economy to a more open, globally integrated market. The landmark economic liberalization of 1991 restructured India's trade and investment landscape, leading to significant change in foreign direct investment (FDI) trends. The liberalization of FDI policies during this period aimed to attract foreign capital, technology, and managerial expertise to fuel industrial development and economic modernization. Prior to 1991, India's FDI regime was highly restrictive, characterized by complex licensing procedures and sectoral barriers that discouraged foreign participation. Post-reform, however, India undertook substantial policy changes, including the relaxation of sectoral caps, simplification of approval processes, and the establishment of Special Economic Zones (SEZs). These reforms collectively improved the investment climate and encouraged sustained foreign inflows.

Figure 1 presents India's FDI inflow data from 1990 to 2022, reflecting a clear upward trend, particularly from the early 2000s onward. During the 1990s, FDI inflows remained modest, indicative of the gradual uptake of liberalization reforms. However, as policy liberalization deepened and investor confidence grew, FDI began to increase significantly. The early 2000s saw India emerge as a hub for business process outsourcing (BPO) and information technology services, attracting substantial investment from multinational corporations. This period was also marked by India's growing consumer base, improved regulatory mechanisms, and active government facilitation of international investments.

The sectoral distribution of FDI in India has also evolved. While early investments were largely confined to traditional sectors such as manufacturing, infrastructure, and telecommunications, the post-liberalization period witnessed a broad diversification into new areas. Sectors such as information technology, automotive, pharmaceuticals, and financial services have become key recipients of FDI, reflecting structural shifts in the Indian economy and aligning with global investment trends. Moreover, the liberalization of retail sector and the advent of initiatives like "Make in India" have further widened the scope of investment opportunities in the country.

FDI has played a vital role in India's economic development by facilitating capital formation, technology transfer, and industrial upgrading. The inflow of foreign capital has helped modernize key sectors and introduce global best practices in production and management. It has also generated employment, contributed to skill development, and integrated India more deeply into global value chains, enhancing the country's competitiveness in international markets.

Given the critical contribution of FDI to economic growth, it is important to assess the institutional and policy frameworks that influence investment decisions. Among these, Double Taxation Treaties (DTTs) represent a crucial instrument for improving investment attractiveness. These treaties aim to eliminate the problem of double taxation, provide legal certainty, and offer a more stable tax environment for foreign investors. By reducing the effective tax burden, DTTs can significantly enhance a host country's appeal to multinational firms. As shown in the work of Blonigen and Davies (2004), DTTs facilitate FDI by removing tax-related barriers and offering fiscal predictability. Additionally, provisions such as exchange of information improve transparency and help combat tax evasion (Donghui et al., 2018). Investors are particularly sensitive to legal certainty and transparency, especially in developing countries where regulatory institutions may be evolving. Studies such as Neumayer (2007), Murthy and Bhasin (2015), Savaş Cevik (2015), Murciego and Laborda (2018), and Kumar and Ansari (2024) affirm the positive impact of DTTs on FDI, citing their role in providing a robust legal and institutional framework to protect investor interests.

Despite the significance of DTTs, limited empirical research has been conducted to evaluate their specific impact on India's bilateral FDI flows using robust econometric methods. Understanding this relationship is essential for policymakers seeking to enhance India's investment environment. Accordingly, this study investigates the determinants of FDI inflows into India from its top 22 investing countries between 1990 and 2022, with a special focus on the role of tax treaties.

2 Literature Review

The relationship between double taxation treaties (DTTs) and foreign direct investment (FDI) has been the subject of extensive research, reflecting varying findings across different contexts and time periods. Blonigen and Davies (2000) conducted one of the earliest and most influential empirical studies examining how double taxation treaties affect foreign direct investment (FDI). Using U.S. data on both inbound and outbound FDI from 1966 to 1992, they first applied a simple treaty dummy variable and found that treaties significantly influenced outbound FDI—but only when using the FDI model later

formalized by Carr, Markusen, and Maskus (2001). They further refined their analysis by introducing the variable "treaty age" to explore whether older treaties had a stronger impact. Their findings revealed that as treaties matured, they had a significantly positive effect on both inbound and outbound FDI flows, suggesting that the benefits of tax treaties grow over time. However, in their subsequent work in 2002 Blonigen and Davies revisited the impact of tax treaties on FDI, this time using data from OECD countries between 1982 and 1992. Their initial OLS results suggested that tax treaties were associated with higher FDI flows and stocks. However, once they distinguished between older treaties and newer ones, they found that the newer treaties had no noticeable impact. When applying fixed effects models, the results were even more striking—indicating that tax treaties might actually reduce FDI. The authors suggested this could be because treaties primarily aim to curb tax evasion and treaty shopping, rather than to directly promote investment.

In a further exploration of tax treaties, Davies (2003) focused on U.S. inbound and outbound FDI from 1966 to 2000, particularly the effects of treaty renegotiations. His results showed that renegotiations had no significant effect on FDI, casting doubt on the view that tax treaties are a key driver of foreign investment. Similarly, Blonigen and Davies (2004) revisited the impact of double taxation treaties (DTTs) on foreign direct investment (FDI), focusing solely on the United States as the source country for FDI. Their study, which spanned from 1980 to 1999, utilized a fixed-effects strategy along with a customized version of the Carr, Markusen, and Maskus (CMM) model. They differentiated between old treaties (signed before 1980) and new treaties (signed after 1980). In their initial analysis, they found a statistically negative effect of new treaties on FDI. However, these initial estimates had undesirable properties, which were corrected by refining the CMM framework to include country-pair fixed effects and a log-linear specification of the variables. With these adjustments, the study found that there was no evidence to support the hypothesis that double taxation treaties significantly affect FDI activity, nor did it find that the positive and negative aspects of treaty formation offset each other.

Contrasting these earlier findings, Egger et al. (2006) examined the impact of tax treaties on outward foreign direct investment (FDI) using a general equilibrium model and a propensity score-matched difference-in-differences estimation strategy. Their analysis of OECD bilateral FDI data from 1985 to 2000 found that newly implemented tax treaties had a significant negative effect on outward FDI stocks. The authors attributed this negative impact to the fact that tax revenues from these treaties were often allocated to public infrastructure, which reduced plant set-up costs and, in turn, decreased the need for outward FDI.

Neumayer (2007) investigated the extent to which double taxation treaties (DTTs) encourage foreign direct investment (FDI) between the United States and developing countries from 1970 to 2001. Using a dyadic analysis with a fixed-effects estimation model, the study found that developing countries with more DTTs signed with major capital-exporting countries, including the U.S., experienced higher overall FDI stocks and greater FDI inflows. Further, when dividing the sample into low-income and middle-income countries, the study concluded that DTTs were only effective in attracting FDI to middle-income countries.

In a separate monadic analysis of OECD DTTs' impact on total inward FDI to developing countries, the study found evidence of a positive effect of DTTs on FDI.

Similarly, Seigmann (2007) analyzed the impact of bilateral tax and investment treaties on FDI. The study employed Gravity and knowledge-capital models to determine the FDI flows from industrialized to developing countries. The study used the panel dataset of 1364 country-pairs over 25 years and concluded that double taxation treaties and investment agreements have a significant positive impact on foreign direct investment flows.

Davies et al. (2009) explored how tax treaties affect multinational firms using micro data from Swedish multinationals. The study found no significant impact of tax treaties on overall affiliate sales. However, it revealed that tax treaties increased the likelihood of Swedish firms investing in a country by establishing affiliates there. Additionally, treaties were found to reduce exports to the parent company while increasing imports of intermediate goods from the parent, suggesting that tax treaties may raise the effective tax rate in the host country. Overall, the study concluded that tax treaties influence certain aspects of multinational behavior but not all dimensions.

Coupe, Orlova, and Skiba (2009) found no evidence that double taxation treaties (DTTs) influence FDI flows between OECD source countries and transitioning economies from 1990 to 2001. Using fixed and random effects in a gravity model, they examined both DTTs and bilateral investment treaties (BITs). The study suggests that while DTTs are expected to have a positive effect on FDI, this effect may be offset by the inclusion of transfer pricing provisions in the treaties, which could have a negative impact on FDI.

Further supporting the positive relationship between tax treaties and FDI, Barthel, Busse, and Neumayer (2010) conducted a dyadic analysis of double taxation treaties (DTTs) and FDI, covering both developed and developing economies from 1978 to 2004. Using a fixed-effects model and Arellano–Bond's Generalised Method of Moments (GMM) for robustness, the study found a positive effect of DTTs on FDI stocks. However, the paper also highlighted potential negative aspects of tax treaties, noting that negotiating these agreements could drain valuable administrative resources, especially in developing countries, and may lead to a loss of revenue for host nations.

BR Parik, J Pankaj, and R Spahr (2011) examined the impact of tax treaties on cross-border portfolio equity flows, valuations, and the cost of capital, using data from 37 host countries and 50 source countries from 1950 to 2008. The study, employing OLS estimation with country and year fixed effects, found that the creation of double taxation treaties (DTTs) increased both foreign direct investment (FDI) and foreign portfolio investment (FPI) by 45%. It also highlighted the significant role of differential corporate and personal tax rates between investing and host countries in influencing cross-border investment flows, with lower-tax countries attracting investment from higher-tax countries. The study concluded that a higher number of DTTs led to increased equity valuations, particularly in developed countries, and a reduction in the cost of equity capital.

Arjan Lejour (2014) investigated the influence of bilateral tax treaties and related tax variables on bilateral FDI flows among OECD member countries from 1985 to 2011. Using OLS estimation with fixed effects and propensity score matching, the study found that newly signed tax treaties, when instrumented with geographic variables, increased bilateral FDI by 21%. The research also highlighted that EU directives on parent-subsidiary relationships significantly boosted FDI stocks. Furthermore, the analysis of treaty shopping revealed that an increase of 20 additional tax treaties could raise bilateral FDI stocks by approximately 50%, and that lower withholding tax rates on dividends in treaties had a notable positive effect on FDI.

Paul L. Baker (2014) examined the effect of tax treaties on FDI using data from 30 OECD home countries and 206 non-OECD host countries between 1991 and 2006. Applying matching econometrics, difference-in-differences estimation, and sensitivity checks, the study found no significant impact of tax treaties on FDI flows. A qualitative review of treaties and domestic tax laws further revealed that many developed countries already offer unilateral relief from double taxation, making treaties redundant and diminishing their influence on multinational investment decisions.

Blonigen, Oldenski, and Sly (2014) examined how specific provisions within bilateral tax treaties—particularly those related to information exchange and coordinated tax treatment—impact U.S. multinational firms across industries. Using firm-level data from 1987 to 2007, they found that in industries relying heavily on differentiated inputs, treaties with Mutual Agreement Procedures (MAP) are more likely to boost affiliate entry and performance. However, they did not find strong average positive effects of treaties on affiliate sales or FDI overall, reinforcing earlier findings that treaty benefits are not uniform across sectors.

Murthy and Bhasin (2015) analyzed the effect of bilateral tax treaties on India's FDI inflows using panel data from 14 countries between 1993 and 2011. Applying a fixed-effects model, they found that tax treaties significantly boost FDI into India. The study also highlighted that both the presence and the age of treaties positively influence FDI. In addition, economic fundamentals such as GDP, per capita income, population, and FDI openness of the investing countries were identified as key determinants of FDI inflows.

Shah and Qayyum (2015) investigated the effect of double taxation treaties on FDI inflows in 15 Latin American and Caribbean developing countries from 1983 to 2013. Their findings suggest that while tax treaties aim to avoid double taxation (which encourages FDI) and curb tax evasion (which may deter FDI), these opposing objectives tend to cancel each other out. As a result, the study found no significant impact of tax treaties on FDI inflows. Instead, factors like market size, development level, trade openness, resource availability, and country-specific conditions played a more influential role in attracting FDI.

Savas Cevik (2015) examined Turkey's outward FDI stocks to 71 host countries between 2001 and 2012, using fixed effects regression analysis. The study found that tax treaties have a positive impact on Turkey's outward FDI, and that older treaties tend to be more effective in promoting investment.

Sunghoon Hong (2017) explored how the global tax treaty network affects FDI by constructing a tax rate matrix among 70 countries and identifying tax-minimizing investment routes. The study found that routes optimized for lower tax rates significantly increase FDI inflows—by over \$4 billion on average compared to non-optimized routes. The findings suggest that aligning direct investment routes with tax-efficient structures can boost FDI flows, further supporting the hypothesis that double taxation treaties facilitate investment in a globalized economy.

Beer and Loeprick (2018) conducted a comprehensive analysis of African countries, focusing on tax treaties with investment hubs over the period 1985 to 2015. Their findings, derived from a Difference-in-Difference approach, suggested that tax treaties did not foster increased investment in these regions. In fact, the study highlighted a significant downside: tax treaties were found to increase the risk of tax revenue loss for source countries. They pointed to the growing phenomenon of treaty shopping, where income is rerouted to jurisdictions with more favorable tax treatment, thus complicating investment patterns. This research provides a more critical perspective, cautioning that DTTs may have unintended negative consequences in certain contexts.

In a similar vein, Murciego and Laborda (2018) explored the impact of DTTs on Spain's inward and outward FDI between 1993 and 2013. Using the knowledge-capital model, they found that DTTs generally had a positive effect on FDI, aligning with the theoretical expectations. However, their analysis also revealed that new treaties with developing countries had a particularly positive effect on outward FDI, whereas inward FDI remained unaffected. This nuanced finding indicates that the influence of tax treaties is not uniform across all countries and highlights the importance of the specific treaty terms and the nature of the countries involved. For Spain, treaties with developed nations appeared to be more beneficial, reinforcing the idea that the benefits of DTTs can vary depending on the investment relationship.

A broader view of FDI determinants is provided by Donghui et al. (2018), who studied the role of trade openness in driving FDI inflows in India, Pakistan, and Iran from 1982 to 2012. Their analysis, using Fixed Effects and Pooled OLS techniques, showed a consistent positive relationship between trade openness and FDI across these countries. While the study didn't directly focus on DTTs, it emphasized that macroeconomic factors like GDP per capita, inflation, and exchange rates also play a significant role in attracting FDI. The findings underscore that, in addition to tax treaties, policies fostering trade openness are key to boosting FDI in developing economies.

Kumar and Millimet (2018) shifted the focus to the United States, examining how bilateral tax treaties impacted U.S. FDI between 1980 and 1999 using quintile regression. Their study uncovered a crucial insight: the effect of tax treaties varies depending on the level of FDI in the host country. For countries with low initial FDI, treaties tended to have a positive impact. However, for countries with high levels of FDI, the impact of treaties was either negative or diminished. This finding suggests that DTTs may not always be a beneficial tool for countries that already have strong investment flows, indicating that treaties may not be a one-size-fits-all solution for encouraging FDI.

Dong (2019) further investigated the ASEAN region, analyzing the effects of DTTs on FDI inflows from 1989 to 2016. His study revealed that while newly negotiated DTTs had little to no effect on FDI, older treaties actually resulted in reduced FDI inflows. This finding challenges the conventional wisdom that DTTs always serve to increase FDI, suggesting that outdated or poorly structured treaties may even deter foreign investment. Dong's research highlights the importance of regularly updating treaties to ensure they remain relevant in a changing global investment environment.

More recently, Kaur, Kumar, and Ansari (2024) studied the determinants of FDI inflows into India from 2000 to 2019, using advanced econometric models like FGLS, PPML, and Newey-West standard errors. Their findings indicated that factors such as FDI openness, gross fixed capital formation, and exports positively influenced FDI inflows. Interestingly, they also found that the GDP per capita of source countries had a negative impact on FDI inflows from certain nations. This suggests that while tax treaties play a role in shaping FDI patterns, other economic variables, including domestic investment policies, play an equally important part in attracting foreign investment into India.

3 Data Description and Methodology

3.1 Theoretical Framework

The gravity model is a widely used tool in international economics to explain trade patterns between countries. Drawing inspiration from Newton's law of gravity, it suggests that trade between two countries is positively related to their economic size (usually measured by GDP) and negatively related to the distance between them. Simply put, larger economies tend to trade more, while greater physical distance often reduces trade due to higher costs.

The gravity model of trade, introduced by Tinbergen in 1962, has evolved significantly over time to incorporate various factors influencing trade, such as trade agreements, common languages, and historical ties. Building on Tinbergen's work, Linnemann (1966) derived the gravity equation using the Walrasian general equilibrium model. Anderson (1979) was the first to offer a theoretical foundation for the model by introducing the concepts of product differentiation and imperfect competition. He argued that the origin of products, as per the Armington assumption, plays a key role in trade patterns, distinguishing between tradable and non-tradable goods.

While Anderson's gravity model did not consider prices, Bergstrand (1985, 1989, 1990) extended it by incorporating price effects through a general equilibrium approach. His work highlighted the role of monopolistic competition and economies of scale. Deardorff (1998) further refined the model, showing that it could be derived from the Heckscher-Ohlin framework under perfect competition. The model's robustness was further enhanced by Feenstra (2004), who introduced advanced econometric techniques to address biases and unobserved heterogeneity, solidifying the gravity model's position as a cornerstone in international trade analysis.

The basic form of the model is:

$$\text{Trade}_{ij} = G \frac{M_i \cdot M_j}{D_{ij}}$$

Where Trade_{ij} is the trade between countries i and j , M_i and M_j are their GDPs, D_{ij} is the distance, and G is a constant.

This model is highly suitable for the present study. It not only captures core economic forces like market size and trade costs but is also flexible enough to include additional variables such as double taxation treaties, historical ties, legal systems, and political factors. These additions allow for a more comprehensive analysis of drivers of India's FDI inflows.

3.2 Data Sources

This study uses secondary data from 22 top investing countries with which India has double taxation agreements (DTTs) from 1990 to 2022. FDI data is sourced from DIPP's FDI Newsletters. Information on India's tax treaties comes from the OECD and Worldwide-tax.com. Other independent variables, such as GDP, trade openness, and FDI openness, are obtained from the World Bank's World Development Indicators (WDI). Data on factors like distance, official language and colonial ties have been collected from the CEPII Gravity Database. The detailed description and source of these variables are given in the Table 1.

Table 1
Description of Variables and Data Sources

Variable	Description	Data Source
DTT	Double Taxation Treaty (Dummy variable) = 1 if India has signed a DTT with the partner countries; 0 otherwise	OECD
FDI	Foreign Direct Investment Inflow in India	SIA Newsletters of DIPP
GDPind	Gross Domestic Product of India	WDI
GDPptr	Gross Domestic Product of partner countries	WDI
TOPind	Trade openness of India	WDI
TOPptr	Trade openness of partner countries	WDI
FDIOP	FDI openness of India	WDI
DIST	Distance from the capital cities of India and its partner countries	CEPII
ComOL	Common Official Language (Dummy variable) = 1 if India and its partner country have a common official language; 0 otherwise	CEPII
ColRel	Colonial Relationship (Dummy variable) = 1 if India and its partner country were in a colonial relationship in the past; 0 otherwise	CEPII
<i>Source: scholars own compilation</i>		

3.3 Sample and Type of Data

This study examines a sample of 22 top investing countries with which India has double taxation treaties (DTTs) for the period from 1990 to 2022. This timeframe is significant as it marks the period of India's liberalization, starting in 1990. The selected countries, including Mauritius, the USA, the UK, France, Russia, and others, account for nearly 80% of India's total trade and investment. Countries like the Cayman Islands and Bermuda were excluded due to the absence of DTTs with India. The study utilizes panel data, combining both cross-sectional and time-series data, which offers advantages such as increased variability, reduced collinearity, and enhanced efficiency (Gujarati, 2004). This approach is ideal for analyzing both time and entity-specific effects, making it well-suited for examining the influence of DTTs on India's trade and FDI flows.

3.4 Variables of the Study and their Measurement

This study examines various dependent and independent variables to explore the impact of tax treaties on FDI inflows to India. The variables have been carefully selected based on theoretical insights and empirical findings from the literature. Monetary variables are expressed in constant 2015 US dollars to adjust for inflation and are measured in logarithmic form to address heteroscedasticity and interpret coefficients as elasticities.

3.4.1 Dependent Variables

- **Foreign Direct Investment (FDI) Inflow:** Defined by the IMF as an investment of at least 10% ownership with managerial control, FDI is crucial for developing economies like India. The study investigates how double taxation treaties (DTTs) affect FDI inflows, as they can play a significant role in attracting foreign investment by reducing barriers like double taxation.

3.4.2 Independent Variables

- **Double Taxation Treaty (DTT):** This binary variable indicates the presence of a DTT between India and its partner country in a given year (1 if present, 0 if absent). These treaties aim to eliminate double taxation and promote trade and investment (Murthy & Bhasin, 2015).
- **Gross Domestic Product (GDP):** GDP measures the total monetary value of all final goods and services produced within a country. It reflects the market size of an economy, with larger economies being more likely to attract investment opportunities (Yusuf et al., 2021; Alfaro et al., 2004).
- **Distance:** Measured as the distance between the capital cities of India and its partner country, this variable highlights the impact of travel, communication, and trade costs. Greater distances are expected to negatively impact FDI inflows.
- **Trade Openness:** This ratio of exports and imports to GDP measures a country's participation in the global trade system. Greater openness is expected to encourage trade and investment.
- **FDI Openness:** The degree to which a country is open to FDI, measured by net FDI inflows as a percentage of GDP. Higher FDI openness is expected to result in more FDI inflows, especially in countries with favourable tax treaties.
- **Common Official Language:** A binary variable that indicates whether India and the partner country share a common official language. A shared language reduces communication barriers, making international investment and business operations easier.
- **Colonial Relationship:** This dummy variable indicates whether the partner country had a colonial relationship with India in the past. A value of 1 indicates a colonial connection, while 0 indicates no such relationship.

The analytical framework of this study is based on the gravity model, initially developed by Tinbergen (1962). The gravity model has also been adapted to study FDI flows, where the volume of FDI between two countries is a function of their economic size and distance. In the context of this study, the gravity model is specified to include variables that capture the economic, geographical, and institutional factors influencing FDI inflows into India. A panel data regression model is applied, in accordance with Gujarati and Porter (2009), Baltagi and Kao (2001), Greene (2003) and Sandeep, Pushp and Mohd Arshad (2024). The augmented gravity model for analysing the impact of tax treaties on FDI is given as:

$$\ln(FDI_{ijt}) = \alpha_0 + \beta_1 \ln(GDP_{it}) + \beta_2 \ln(GDP_{jt}) + \beta_3 \ln(Dist_{ij}) + \beta_4 (DTT_{ijt}) + \beta_5 (ComB_{ij}) + \beta_6 (ComOL_{ij}) + \beta_7 \ln(ColRel_{ij}) + \beta_8 (ComCol_{ij}) + \epsilon_{ijt}$$

3.5 Methodology

3.5.1 PPML Econometric approach

The study employs the Poisson Pseudo Maximum Likelihood (PPML) estimator to examine the impact of India's tax treaties and various other determinants on India's FDI inflow. Traditional Ordinary Least Squares (OLS) regressions can yield biased and inconsistent estimates when there is heteroskedasticity or when the dependent variable (FDI inflows) includes zero values, which is common in FDI data. The PPML model handles these issues by providing robust and consistent estimations even in the presence of zeros and heteroskedasticity. Several studies have validated the use of PPML in analysing FDI and trade flows. For instance, Camarero et al. (2020) employed the PPML model to study the determinants of FDI in Spanish regions, and Nguyen et al. (2020) used it to examine bilateral FDI determinants among Asian countries. These applications highlight the model's robustness and reliability in handling data peculiarities in trade and FDI studies. Additionally, Silva and Tenreyro (2006) highlighted that when PPML coefficients are computed they are typically smaller and more accurate compared to the Ordinary Least Square (OLS) coefficients. Head and Mayer (2014) further supported the use of PPML model due to its advantages in handling dummy variables over other models. In the context of this objective PPML model takes the following form.

$$\ln FDI_{ijt} = \exp\{\alpha_0 + \alpha_1(DTT_{ijt}) + \alpha_2 \ln(GDP_{it}) + \alpha_3 \ln(GDP_{jt}) + \alpha_4 \ln(TOP_{it}) + \alpha_5 \ln(TOP_{jt}) + \alpha_6 \ln(FDIOP_{it}) + \alpha_7 \ln(Dist_{ij}) + \alpha_8(ComOL_{ij}) + \alpha_9(ColRel_{ij}) + \eta_i + \nu_t + \varepsilon_{ijt}\}$$

Where: $\ln FDI_{ijt}$ is the natural log of FDI inflow from country i to country j at time t . DTT_{ijt} is the dummy variable indicating the presence or absence of a double taxation treaty between India and its partner country in time t . $\ln(GDP_{it})$ and $\ln(GDP_{jt})$ are the natural logs of GDP of the host country and investing countries. $\ln(TOP_{it})$ and $\ln(TOP_{jt})$ are the natural logs of Trade Openness of India and investing countries. $\ln(FDIOP_{it})$ is the natural log of FDI openness of India. $\ln(Dist_{ij})$ is the natural log of the distance between India and its partner countries. $ComOL_{ij}$ is a dummy variable indicating common official language. $ColRel_{ij}$ is a dummy variable representing colonial relationship between India and its treaty partner country. η_i and ν_t are country and time fixed effects. ε_{ijt} is the error term.

3.5.2 Robustness Checking

For checking the robustness of the results the study employs feasible generalized least square (FGLS) and Newey-West standard error model. FGLS is particularly effective in dealing with issues of heteroscedasticity and autocorrelation, providing more efficient estimates compared to Ordinary Least Squares (OLS) in such contexts. Similarly The Newey-West estimator corrects the standard errors of OLS

estimates, making them robust to serial correlation, heteroscedasticity and cross-sectional dependence. FGLS is widely used in FDI literature due to its efficiency in handling heteroscedasticity and autocorrelation (Wei et al., 2020; Shah et al., 2020; Haiyue and Manzoor, 2020; Huynh, 2020). Similarly Studies such as Kim (2010) and Sandeep, Pushp and Mohd Arshad (2024) have demonstrated the effectiveness of the Newey-West standard error model in providing robust inference when analysing the determinants of FDI inflow.

By employing the PPML, FGLS, and Newey-West standard error models, this study ensures that the results of the analysis are robust and reliable. The use of these models addresses various econometric issues, providing a comprehensive understanding of how tax treaties influence FDI decisions. The combination of these methodologies offers a nuanced approach, accounting for potential biases and ensuring the validity of the results.

4 Results and discussion

4.1 Summary Statistics

An essential first step in panel data analysis is the use of descriptive statistics, which provide an understanding of the properties of the data and facilitate the interpretation and application of findings. Descriptive statistics are a collection of measurements of two things: location and variability. The location of a variable indicates its central value, which is typically represented by the mean. Variability is the spread of data from the central value, often known as variance or standard deviation. The results of the descriptive statistics are presented in the Table 2.

Table 2
Descriptive Statistics

Stats	Mean	Min	Max	SD	Skewness	Kurtosis	N
L_FDI	7.6337	0.0023	14.1416	3.3256	-0.4228	2.4129	726
DTT	0.6956	0	1	0.4605	-0.8501	1.7227	726
L_GDPind	27.7894	26.8658	28.7167	0.5851	-0.0018	1.6968	726
L_GDPptr	27.102	22.1181	30.6721	1.7932	-0.6392	3.3023	726
L_TOPind	3.5186	2.7412	4.0217	0.3846	-0.4464	1.8314	726
L_TOPptr	4.3685	2.7551	6.0807	0.6998	0.2454	2.765	726
L_DIST	8.5687	7.7463	9.3977	0.3327	-0.1969	3.9788	726
L_FDIOP	1.2655	0.0272	3.6205	0.824	0.5787	3.1628	726
ComOL	0.1818	0	1	0.386	1.6499	3.7222	726
ColRel	0.0455	0	1	0.2084	4.3644	20.0476	726

Source: Scholar's computation From Stata.

Descriptive statistics provide an overview of the dataset's distribution and variability. Table 2 summarizes key variables such as FDI inflows (L_FDI), tax treaties (DTT), GDP, trade openness, and other control variables. FDI inflows exhibit moderate variability (CV = 0.44) with a near-symmetric distribution. The presence of tax treaties is high (mean = 0.70), with notable variability due to their binary nature. India's GDP (L_GDPind) shows low variability and a symmetric distribution, while partner countries' GDP (L_GDPptr) displays greater dispersion and a left-skewed distribution. Trade openness of India (L_TOPind) has low variability, while that of partner countries (L_TOPptr) is moderately variable and slightly right-skewed. Distance (L_DIST) has very low variability, reflecting its constant nature across country pairs. FDI openness (L_FDIOP) shows higher variability, with a near-normal distribution. Binary variables like Common Official Language (ComOL) and Colonial Relationship (ColRel) have high coefficients of variation due to their sparse values, indicating limited prevalence across the dataset.

4.2 Correlational Analysis

Table 3 presents the correlation matrix to assess linear relationships. FDI inflows (L_FDI) are positively correlated with tax treaties (0.624), suggesting treaties play a significant role in attracting FDI by reducing tax uncertainty. India's GDP (0.681) and trade openness (0.624) also show strong positive associations with FDI, reflecting economic scale and market integration as key drivers.

Table 3
Correlation Matrix

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) L_FDI	1.000									
(2) DTT	0.624	1.000								
(3) L_GDPind	0.681	0.526	1.000							
(4) L_GDPptr	0.102	0.282	0.158	1.000						
(5) L_TOPind	0.624	0.487	0.857	0.143	1.000					
(6) L_TOPptr	0.164	-0.013	0.176	-0.663	0.191	1.000				
(7) L_DIST	0.275	0.195	0.000	0.339	0.000	-0.265	1.000			
(8) L_FDIOP	0.618	0.504	0.717	0.120	0.809	0.153	0.000	1.000		
(9) ComOL	0.390	0.219	0.000	-0.051	0.000	0.060	0.333	0.000	1.000	
(10) ColRel	0.123	0.073	0.000	0.174	0.000	-0.112	0.160	0.000	0.463	1.000

Source: Scholar's computation From Stata.

Variables like partner countries' GDP and trade openness show weaker correlations with FDI, implying limited direct influence. Distance, despite being weakly positive, shows a reduced role due to globalization. The correlation matrix also indicate no severe multicollinearity, supporting the reliability of subsequent regression analysis.

4.3 Diagnostic Tests

To ensure the validity of regression results, several diagnostic tests were conducted. The Variance Inflation Factor (VIF) analysis (Table 4) revealed no severe multicollinearity, with all VIFs below 10 and a mean VIF of 2.55, indicating reliable coefficient estimates. The Wooldridge test detected first-order autocorrelation, while the Breusch-Pagan/Cook-Weisberg test confirmed heteroskedasticity—both justifying the use of robust standard errors and estimators such as PPML, FGLS, and Newey-West for consistent inference. Cross-sectional dependence was not found, as indicated by the Breusch-Pagan LM test, validating the assumption of independent units. The results of these tests are shown in the Table 5. Lastly, the Levin-Lin-Chu panel unit root test (Table 6) showed all variables to be stationary at level, with p-values < 0.05, confirming that the data met stationarity requirements. These diagnostics support the use of PPML with robust standard errors as the primary estimation technique for this study.

Table 4
VIF

Variable	VIF	1/VIF
L_TOPind	5.44	0.184
L_GDPind	4.08	0.2451
L_FDIOP	2.92	0.3428
L_GDPptr	2.4	0.416
L_TOPptr	2.13	0.4684
DTT	1.62	0.6156
ComOL	1.62	0.619
ColRel	1.37	0.7324
L_DIST	1.34	0.7481
Mean VIF	2.55	

Source: Scholar's computation From Stata.

Table 5
Diagnostic Tests

Test	Null Hypothesis	Test Statistic	p-value	Decision
Wooldridge Test for Autocorrelation	No first-order autocorrelation	$F(1, 21) = 12.476$	0.0020	Reject H_0 (Autocorrelation present)
Breusch–Pagan/Cook–Weisberg Test	Constant variance (Homoscedasticity)	$\chi^2(1) = 20.97$	0.0000	Reject H_0 (Heteroskedasticity present)
Breusch-Pagan LM Test for Cross-sectional Dependence	No cross-sectional dependence	$\chi^2(231) = 743.212$	0.2573	Fail to reject H_0 (No dependence)
<i>Source: Scholar's computation using Stata</i>				

Table 6
Levin-Lin-Chu panel unit root test

Variables	Unadjusted t	Adjusted t	P-value	Decision
L_FDI	-9.5284	-6.6098	0.0000	Stationary (Reject H0)
L_GDPind	-2.9931	-2.7868	0.0027	Stationary (Reject H0)
L_GDPptr	-6.6100	-5.9560	0.0000	Stationary (Reject H0)
L_TOPind	-8.4893	-4.9179	0.0000	Stationary (Reject H0)
L_TOPptr	-6.4284	-2.9724	0.0015	Stationary (Reject H0)
L_FDIOP	-9.9077	-5.6264	0.0000	Stationary (Reject H0)
DTT	-7.7485	-2.0848	0.0185	Stationary (Reject H0)

Source: Scholar's computation From Stata.

4.4 Estimated results of PPML model

This study employs the Poisson Pseudo Maximum Likelihood (PPML) estimator to examine how India's double taxation treaties (DTTs) and other key variables influence foreign direct investment (FDI) inflows. The PPML method, as advocated by Silva and Tenreyro (2006), is particularly well-suited for this analysis because it handles zero FDI values and corrects for heteroskedasticity—two common challenges in FDI data. Unlike traditional OLS, which can produce biased results under such conditions, PPML yields consistent and robust estimates.

Several studies have validated the use of PPML in analysing FDI and trade flows. For instance, Camarero et al. (2020) employed the PPML model to study the determinants of FDI in Spanish regions, and Nguyen et al. (2020) used it to examine bilateral FDI determinants among Asian countries. These applications highlight the model's robustness and reliability in handling data peculiarities in trade and FDI studies.

Moreover, the independent variables used in our study, such as tax treaties, common official languages and colonial relationship are dummy variables indicating the presence or absence of each factor to predict their impact on India's FDI inflow. Following Silva and Tenreyro (2006), the effect of change in variable x on variable y is calculated by $\{(e^{\alpha} - 1) \times 100\}$. Where α is the coefficient of a dummy variable. Interpreting the coefficients of dummy variables in an exponential form is essential in log-linear models, such as the gravity model of trade, to provide meaningful percentage changes (Baier and Bergstrand 2007 and Head and Mayer 2014). This method translates the raw coefficients into a comprehensible format by using this formula. The results of the PPML model are reported in the Table 7.

Table 7
Results of gravity model: PPML estimation

L_FDI	Coefficient	Robust Std. Err.	Z	P Value
DTT	0.3458005	0.0385397	8.97	0.000
L_GDPind	0.2716534	0.0246362	11.03	0.000
L_GDPptr	-0.0127102	0.0063716	-1.99	0.046
L_TOPind	0.1457194	0.0541965	2.69	0.007
L_TOPptr	0.0505835	0.0177975	2.84	0.004
L_FDIOP	0.0598059	0.0159652	3.75	0.000
L_DIST	0.2305401	0.0353176	6.53	0.000
ComOL	0.2589404	0.0181738	14.25	0.000
ColRel	-0.0457985	0.0244002	-1.88	0.041
Constant	-8.317454	0.6979761	-11.92	0.000
N	726			
R-square	0.672121			

Source: Scholar's computation From Stata.

The empirical results of PPML reported in the Table 7 shows that tax treaty is statistically significant and positive at 1% level of significance. This implies that the presence of a double taxation treaty (DTT) significantly increases FDI inflows into India by approximately $\{(e^{0.3458005}-1) \times 100\}$ 41.31% compared to those without a tax treaty. This confirms the hypothesis that DTTs reduce tax uncertainties and encourage international investment in India. These findings align with Blonigen and Davies (2004), who noted that DTTs help attract FDI by mitigating double taxation and offering legal certainty.

The sign of the coefficient of India's GDP is positive and significant at 1% level of significance. The coefficient of GDP of India states that a 1% increase in GDP of India is associated with 0.27% increase in India's FDI inflow. This positive relationship suggests that higher economic growth in India is a major attractor of FDI, consistent with the theory that larger economies offer more opportunities and a larger market for investment (Alfaro et al., 2004). The coefficient for the GDP of partner countries is negative and significant at the 5% level, implying that a 1% increase in GDP of partner country is associated with a decrease of 0.012% in FDI inflow. One possible explanation is that wealthier countries might have more profitable domestic investment opportunities, reducing their need to invest abroad. This finding is consistent with the investment diversion hypothesis posited by Helpman (1984), where increased domestic opportunities lead to reduced outbound FDI.

Regarding the impact of trade openness, it is observed that coefficient for India's trade openness is positive and statistically significant at 1% level. This implies that if trade openness increases by 1%, FDI inflow of India will increase by approximately 0.15%, highlighting the role of trade liberalization in attracting foreign investment (Edwards, 1998). The sign of the coefficient of trade openness of partner countries is positive and significant at 1% significance level. This positive relationship indicates that higher trade openness in partner countries facilitates FDI into India, supporting the argument that liberal trade policies can enhance investment flows (Busse and Hefeker, 2007). The result exhibits that for a 1% increase in trade openness of partner countries, a boost of 0.05% FDI inflow will occur to India

The sign of coefficient of FDI openness of India is positive and statically significant at 1% level of significance. This implies that if the FDI openness is increased by 1% consequently, FDI inflow will increase approximately by 0.06%. Therefore FDI openness leads to more FDI inflows. This is in line with the study of Asiedu, (2002) which highlights the importance of a liberal FDI regime in attracting foreign investments.

Contrary to the traditional gravity model expectation, the coefficient for distance is positive and statistically significant at 1%. The coefficient of the log of distance indicates that a 1% increase in distance between India and its treaty partner, leads to an increase in FDI inflow by around 0.23%. This suggests that greater distance between India and its partner countries is associated with higher FDI inflows. One explanation could be the nature of modern trade and investment, where technological advancements and globalization reduce the impact of geographical distance on investment decisions (Disdier and Head, 2008). Another explanation as noted by Kayam and Hisarciklilar (2009) could be the effect of distance on FDI varies by type of FDI. For horizontal FDI, aimed at market expansion, greater distance may promote investment, whereas vertical FDI, seeking production efficiency, prefers lesser distance (Bhasin and Manocha, 2016). The positive impact of distance on FDI inflows into India in our study suggests that foreign investors are inclined towards market-seeking (horizontal) FDI, viewing India as a strategic market for expansion despite the geographical distance.

The coefficient of common official language is positive and statically significant at 1% level of significance, which implies that having a common official language between India and its treaty partner countries increases FDI inflows by approximately $\{(e^{0.2589404}-1) \times 100\}$ 29.55% compared to those countries not have a common official language. This positive effect indicates that common language reduces communication barriers and transaction costs, thus facilitating international business operations. This finding is consistent with Ghemawat (2001), who highlighted the significance of shared language in enhancing cross-border economic activities. The coefficient of colonial relationship is negative and significant at 5% level of significance, which implies that presence of a colonial relationship decreases FDI inflows by approximately $\{(e^{-0.0457985}-1) \times 100\}$ 4.47%. This negative relationship suggests that historical colonial ties may negatively impact current FDI inflows, potentially due to lingering negative perceptions and mistrust stemming from colonial history. This result is supported by Head et al. (2010), who argued that past colonial relationships could result in adverse economic perceptions that persist over time. Moreover, the model has an R-squared value of 0.672121 which

indicates that approximately 67.21% of the variation in FDI inflows is explained by the independent variables in the model. This suggests a good fit of the model to the data.

The PPML results reveal that double taxation treaties significantly enhance FDI inflows into India, reinforcing the importance of international tax cooperation in promoting investment. Additionally, economic fundamentals such as GDP and trade liberalization remain key drivers of FDI. Institutional variables, including language and historical ties, also play a meaningful role. Surprisingly, geographic distance no longer hinders investment, reflecting a shift in how modern investors view market access. These findings underscore the multifaceted nature of FDI determinants and offer important implications for policymakers aiming to enhance FDI inflows through targeted economic and institutional reforms.

4.5 Robustness Checking

To ensure the robustness of the results, this study further examined the long-run relationship between FDI inflows and their determinants using the Feasible Generalized Least Squares (FGLS) method and the Newey-West standard error model. These methods were selected to address potential issues of heteroskedasticity and serial correlation that could undermine the validity of regression estimates. Prior studies—such as Wei et al. (2020), Shah et al. (2020), and Kaur et al. (2024)—have demonstrated the effectiveness of FGLS in providing robust estimates in FDI-related research. Similarly, Kim (2010) employed the Newey-West approach to account for autocorrelation and heteroskedasticity in FDI modelling, establishing its relevance for the current analysis.

Table 8
Feasible Generalized Least Square Model

L_FDI	Coefficient	Std. Err.	Z	P value
DTT	1.653487	0.204715	8.08	0.000
L_GDPind	2.550921	0.274874	9.28	0.000
L_GDPptr	-0.171821	0.073375	-2.34	0.019
L_TOPind	0.761945	0.380935	2.00	0.045
L_TOPptr	0.002224	0.196427	0.01	0.991
L_FDIOP	0.412880	0.074823	5.52	0.000
L_DIST	1.197706	0.434277	2.76	0.006
ComOL	2.345986	0.296551	7.91	0.000
ColRel	-0.741300	0.524528	-1.41	0.048
Constant	-73.345370	7.761389	-9.45	0.000

Source: Scholar’s computation From Stata.

Table 9
Newey-West Standard Error Regression Model

L_FDI	Coefficient	Newey–West Std. Err.	t	P Value
DTT	1.858577	0.210552	8.83	0.000
L_GDPind	2.519222	0.224663	11.21	0.000
L_GDPptr	-0.107712	0.050700	-2.12	0.034
L_TOPind	0.209954	0.428860	0.49	0.025
L_TOPptr	0.243341	0.134784	1.81	0.671
L_FDIOP	0.461203	0.137256	3.36	0.001
L_DIST	1.674284	0.238732	7.01	0.000
ComOL	2.504012	0.156003	16.05	0.000
ColRel	-0.658816	0.187264	-3.52	0.000
Constant	-77.904720	5.764531	-13.51	0.000

Source: Scholar's computation From Stata.

The results of the FGLS model (Table 8) indicate that the coefficient of tax treaty is positive and statistically significant at the 1% level. In the context of the gravity model the coefficient of tax treaty (dummy variable) implies that the presence of a tax treaty is associated with an approximately $\{(e^{1.653487}-1) \times 100\}$ 422.5% increase in FDI inflows to India. These results are consistent with the findings from the PPML model. Similarly the coefficients of GDP of India, FDI openness of India, and distance are positive and statistically significant at 1% level of significance, implying that a 1% rise in these variables enhances the FDI inflow by a magnitude of 2.55, 0.41 and 1.20% respectively. Moreover the coefficients of GDP of partner countries and colonial relationship are negative and significant at 5% level. The coefficients of these variables signify that a 1% increase in GDP of partner countries leads to a decrease of 0.17% in FDI inflow and the presence of colonial relationship leads to a decrease of 52.35% in FDI inflow. The coefficient of trade openness of India is positive and statistically significant at 5%, suggesting that 1% increase in trade openness of India leads to 0.76% increase in FDI inflow in India. The impact of common official language which is a dummy variable is positive and statistically significant at 1% level of significance. However, the variable trade openness of partner countries is having an insignificant relationship with India's FDI inflow in FGLS model.

The study further strengthened the robustness of the results by obtaining long-run elasticity coefficients by employing the Newey-West standard error regression model. The results of this model are presenting in the Table 9. The empirical outcomes derived from this method indicate that the results of this model are consistent with those obtained from both the FGLS and the primary PPML estimator. Therefore there is no need to again discuss these results as they are similar to those discussed in FGLS and PPML model. Key determinants such as the presence of a tax treaty, GDP of India, trade openness of India, FDI

openness of India, distance and common official language are consistently positive and significant across all models, reinforcing their importance in influencing FDI inflows into India. These robustness checks validate the results obtained from the PPML model, enhancing the reliability of the study's conclusions regarding the impact of double taxation treaties and other determinants on FDI inflows to India. The empirical results presented in FGLS model and Newey-West model corroborate those found in PPML model, indicating consistency across different methodological approaches.

5 Conclusion

This study investigated the impact of Double Taxation Treaties (DTTs) on Foreign Direct Investment (FDI) inflows into India, using panel data from the country's top 22 investing partners between 1990 and 2022. The analysis employed the Poisson Pseudo Maximum Likelihood (PPML) model as the primary estimation technique, supplemented by Feasible Generalized Least Squares (FGLS) and Newey-West standard error models to ensure robustness.

Preliminary analysis, including summary statistics and correlation matrices, provided basic insights into the dataset's structure, while diagnostic tests revealed heteroscedasticity and autocorrelation—common features of panel data—which were addressed using appropriate econometric methods.

The regression analysis included different models to analyse the influence of DTTs on FDI inflows. The Poisson Pseudo Maximum Likelihood (PPML) model was chosen for its tolerance to heteroscedasticity and its ability to handle zero FDI inflows effectively. The results from the PPML model demonstrated the significant positive impact of DTTs on FDI inflows into India. This finding was consistent across the alternative models (FGLS and Newey-West standard error models), which were applied to check the robustness of the results. The significant positive coefficients of the DTT variable across all models validated the notion that double taxation treaties play a major role in promoting FDI inflows by removing tax barriers and providing a predictable and stable tax environment for foreign investors. The consistency across these models enhances the credibility of the findings, demonstrating that the observed relationships are not driven by specific econometric issues such as heteroscedasticity or autocorrelation. The findings offer robust evidence that such treaties were beneficial for creating a stable and attractive investment environment in India during the sample period.

The findings of this study carry important implications for both policymakers and future researchers. The consistent and significant positive impact of Double Taxation Treaties (DTTs) on FDI inflows into India suggests that such treaties are more than just legal instruments—they serve as vital tools for enhancing investor confidence by offering tax certainty and reducing fiscal barriers. This underscores the need for Indian policymakers to not only maintain but also strategically expand and modernize the country's network of tax treaties, particularly with emerging markets and major capital-exporting nations. Alongside treaty reforms, strengthening domestic economic fundamentals—such as improving ease of doing business, ensuring regulatory transparency, and maintaining macroeconomic stability—will further complement the benefits of DTTs. Looking ahead, future research could benefit from a more granular approach by examining the sector-specific effects of tax treaties, assessing the role of specific treaty

provisions, and exploring firm-level data to understand how investors respond to treaty-driven incentives across different industries and country contexts.

Declarations

Ethical Approval and Consent to Participate

Not applicable. This study does not involve any human participants or animal subjects.

Consent for Publication

Not applicable. This manuscript does not contain any individual person's data in any form.

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Author Contribution

Firdous Ahmad Hurrah conceptualized the study, conducted the data analysis, and wrote the main manuscript text. Khalid Ashraf Chisti provided research guidance, supervised the analytical framework, and reviewed the manuscript. All authors approved the final version of the manuscript. The author gratefully acknowledges the academic guidance and supervision provided by Dr. Khalid Ashraf Chisti in the completion of this research. The author did not receive any external funding for this research.

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Data Availability

The data used in this study is available from publicly accessible databases and can be shared upon reasonable request.

References

1. Alfaro, L., Chanda, A., Kalemli-Ozcan, S., & Sayek, S. (2004). FDI and economic growth: the role of local financial markets. *Journal of International Economics*, 64(1), 89–112.
2. Anderson, J. E. (1979). A theoretical foundation for the gravity equation. *The American Economic Review*, 69(1), 106–116.

3. Asiedu, E. (2002). On the determinants of foreign direct investment to developing countries: Is Africa different? *World Development*, 30(1), 107–119.
4. Baier, S. L., & Bergstrand, J. H. (2007). Do free trade agreements actually increase members' international trade? *Journal of International Economics*, 71(1), 72–95.
5. Baker, P. L. (2014). An analysis of double taxation treaties and their effect on foreign direct investment. *International Journal of the Economics of Business*, 21(3), 341–377.
6. Baltagi BH, Kao C (2001) Nonstationary panels, cointegration in panels and dynamic panels: a survey. In: Baltagi BH, Fomby TB, Carter Hill R (eds) *Nonstationary panels, panel cointegration, and dynamic panels*. Emerald Group Publishing Limited, Bingley, pp 7–51
7. Barthel, F., Busse, M., & Neumayer, E. (2010). The impact of double taxation treaties on foreign direct investment: evidence from large dyadic panel data. *Contemporary Economic Policy*, 28(3), 366–377.
8. Beer, S., & Loeprick, J. (2018, January). The Cost and Benefits of Tax Treaties with Investment Hubs. In *Proceedings. Annual Conference on Taxation and Minutes of the Annual Meeting of the National Tax Association* (Vol. 111, pp. 1–36). National Tax Association.
9. Bergstrand, J.H (1989). The generalized gravity equation, monopolistic competition, and the factor-proportions theory in international trade. *Rev Econ Stat* 71(1):143–153
10. Bergstrand, J.H (1990) The Heckscher-Ohlin-Samuelson model, the Linder hypothesis and the determinants of bilateral intra-industry trade. *Econ J* 100(403):1216–1229
11. Bergstrand, J.H. (1985). "The Gravity Equation in International Trade: Some Microeconomic Foundations and Empirical Evidence". *The Review of Economics and Statistics*, 67(3), 474–481.
12. Bhasin, N., & Manocha, R. (2016). Do bilateral investment treaties promote FDI inflows? Evidence from India. *Vikalpa*, 41(4), 275–287.
13. Blonigen, B. A., & Davies, R. B. (2002). Do bilateral tax treaties promote foreign direct investment? In *Advances in International Economics and Economic Growth*. Elsevier.
14. Blonigen, B. A., & Davies, R. B. (2004). The effects of bilateral tax treaties on U.S. FDI activity. *International Tax and Public Finance*, 11(5), 601–622.
15. Blonigen, B. A., Oldenski, L., & Sly, N. (2014). The differential effects of bilateral tax treaties. *American Economic Journal: Economic Policy*, 6(2), 1–18.
16. Blonigen, Bruce A., and Ronald B. Davies. 2000. The Effects of Bilateral Tax Treaties on US FDI Activity. National Bureau of Economic Research Working Paper No. 7929 Cambridge, MA: National Bureau of Economic Research.
17. Busse, M., & Hefeker, C. (2007). Political risk, institutions and foreign direct investment. *European journal of political economy*, 23(2), 397–415.
18. Camarero M, Montolio L, Tamarit C (2020) Determinants of FDI for Spanish regions: evidence using stock data. *Empir Econ* 59(6):2779–2820.
19. Camarero, M., Castillo, J., Picazo-Tadeo, A. J., & Tamarit, C. (2020). Determinants of foreign direct investment in Spanish regions: Evidence from a spatial panel data model. *International Review of Economics & Finance*, 67, 1–12.

20. Castillo-Murciego, Á., & López-Laborda, J. (2018). The effect of double taxation treaties and territorial tax systems on foreign direct investment: Evidence for Spain (Economics Discussion Papers, No 2018-21). Kiel Institute for the World Economy. <http://www.economics-ejournal.org/economics/discussionpapers/2018-2>, 1.
21. Cevik, S. (2015). The Impact of Double Tax Treaties on Foreign Direct Investments: Evidence from Turkey? s Outward FDI's (No. 2504045). International Institute of Social and Economic Sciences.
22. Coupé, T., Orlova, I., & Skiba, A. (2009). The effect of tax and investment treaties on bilateral FDI flows to transition countries. In 9th annual global development conference, Brisbane (Vol. 29, p. 98).
23. Davies, R. B. (2003). Tax treaties, renegotiations, and foreign direct investment. *Economic Analysis and Policy*, 33(2), 251–273.
24. Davies, R. B., Norbäck, P. J., & Tekin-Koru, A. (2009). The effect of tax treaties on multinational firms: New evidence from microdata. *World Economy*, 32(1), 77–110.
25. Deardorff AV (1998) Determinants of bilateral trade: Does gravity work in a neoclassical world? In: Frankel JA (ed) *The regionalization of the world economy*. University of Chicago Press, pp 7–32
26. Disdier, A. C., & Head, K. (2008). The puzzling persistence of the distance effect on bilateral trade. *The Review of Economics and Statistics*, 90(1), 37–48.
27. Dong, Y. (2019). The impact of double tax treaties on inward FDI in ASEAN countries. Singapore Management University School of Accountancy Research Paper, 7(1).
28. Donghui, Z., Yasin, G., Zaman, S., & Imran, M. (2018). Trade openness and FDI inflows: A comparative study of Asian countries. *European Online Journal of Natural and Social Sciences*, 7(2), pp-386.
29. Edwards, S. (1998). Openness, productivity and growth: what do we really know? *The Economic Journal*, 108(447), 383–398.
30. Egger, P., Larch, M., Pfaffermayr, M., & Winner, H. (2006). The impact of endogenous tax treaties on foreign direct investment: theory and evidence. *Canadian Journal of Economics/Revue canadienne d'économie*, 39(3), 901–931.
31. Feenstra, R. C., Markusen, J. R., & Rose, A. K. (2001). Using the gravity equation to differentiate among alternative theories of trade. *Canadian Journal of Economics/Revue canadienne d'économie*, 34(2), 430–447.
32. Feenstra, Robert, 2004. *Advanced International Trade*. Princeton University Press.
33. Ghemawat, P. (2001). Distance still matters: The hard reality of global expansion. *Harvard Business Review*, 79(8), 137–147.
34. Greene, W. H. (2003). *Econometric Analysis* (5th ed.). Pearson Education.
35. Gujarati DN, Porter DC (2009) *Basic econometrics*. McGraw-hill
36. Haiyue L, Manzoor A (2020) The impact of OFDI on the performance of Chinese firms along the 'Belt and Road.' *Appl Econ* 52(11):1219–1239.
37. Head K & Mayer T (2014). Gravity equations: Workhorse, toolkit, and cookbook. In *Handbook of International Economics* (Vol. 4, pp. 131–195). Elsevier.

38. Head, K., Mayer, T., & Ries, J. (2010). The erosion of colonial trade linkages after independence. *Journal of International Economics*, 81(1), 1–14.
39. Helpman, E. (1984). A simple theory of international trade with multinational corporations. *Journal of Political Economy*, 92(3), 451–471.
40. Hong, S. (2018). Tax treaties and foreign direct investment: A network approach. *International Tax and Public Finance*, 25, 1277–1320.
41. Huynh CM (2020). How does the impact of foreign direct investment on institutional quality depend on the underground economy? *Journal of Sustainable Finance & Investment*, 1–16.
42. Kaur, S., Kumar, P., & Ansari, M. A. (2024). An analysis of Indian FDI inflows through an augmented gravity model: exploring new insights. *International Economics and Economic Policy*, 21(2), 435–455.
43. Kayam, S. S., & Hisarciklilar, M. (2009). Revisiting the trade-promoting effect of preferential trade agreements. *The World Economy*, 32(7), 1104–1126. <https://doi.org/10.1111/j.1467-9701.2009.01199.x>
44. Kim, W. (2010). The effect of corporate governance on foreign direct investment. *International Business Review*, 19(3), 227–238.
45. Kumas, A., & Millimet, D. L. (2018). Reassessing the effects of bilateral tax treaties on US FDI activity. *Journal of Economics and Finance*, 42, 451–470.
46. Lejour, A. (2014). The foreign investment effects of tax treaties. CPB Netherlands Bureau for Economic Analysis Discussion Paper, 265.
47. Linnemann H (1966) An econometric study of international trade flows, vol 234. North-Holland Publishing Company, Amsterdam
48. Murthy, K. B., & Bhasin, N. (2015). The impact of bilateral tax treaties: A multi-country analysis of FDI inflows into India. *The Journal of International Trade & Economic Development*, 24(6), 751–766.
49. Neumayer, E. (2007). Do double taxation treaties increase foreign direct investment to developing countries? *The Journal of Development Studies*, 43(8), 1501–1519
50. Nguyen, A. T., Haug, A. A., Owen, P. D., & Genç, M. (2020). What drives bilateral foreign direct investment among Asian economies? *Economic Modelling*, 93, 125–141.
51. Parikh, B. R., Pankaj, J., & Spahr, R. (2011). The impact of double taxation treaties on cross border equity flows, valuations and cost of capital.
52. Santos Silva, J. M. C., & Tenreyro, S. (2006). "The Log of Gravity." *The Review of Economics and Statistics*, 88(4), 641–658.
53. Shah SH, Ameer W, Delpachitra S (2020) OFDI impact on private investment in the Gulf economies. *Sustainability* 12(11):4492.
54. Shah, M. H., & Qayyum, S. (2015). Impact of double taxation treaties on inward FDI in Latin American and Caribbean developing countries. *Business & Economic Review*, 7(1), 1–18.
55. Siegmann, T. (2007). The impact of bilateral investment treaties and double taxation treaties on foreign direct investments. U. of St. Gallen Law & Economics Working Paper, (2008-22).

56. Tinbergen J (1962) Shaping the world economy suggestions for an international economic policy. The Twentieth Century Fund, New York

57. Wei, Y., Xie, E., & Zhang, H. (2020). How do institutional quality and foreign direct investment impact economic growth? Evidence from Asia. Economic Modelling, 90, 60–71.

58. Yusuf, H. A., Afolabi, L. O., Shittu, W. O., Gold, K. L., & Muhammad, M. (2021). Institutional quality and trade flow: Empirical evidence from Malaysia and other OIC member Countries in Africa. Insight on Africa, 13(2), 177–191.

Figures

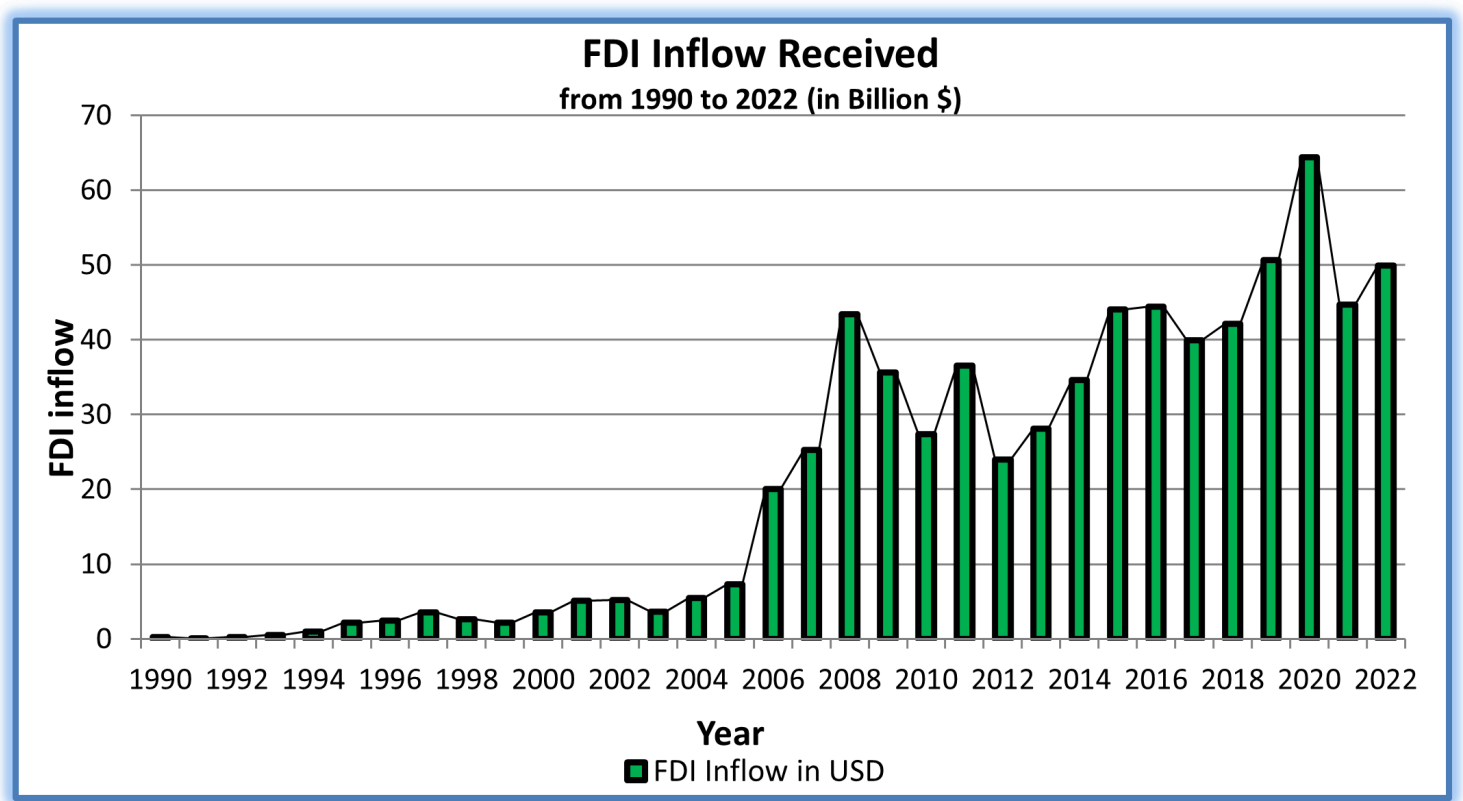


Figure 1

FDI Inflow of India

Source: Author's own calculation by accessing DOTS database.